

YEAR
5

2002

THE UNIVERSITY OF
NEW SOUTH WALES



EDUCATIONAL
TESTING CENTRE



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12/12
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AUSTRALIAN SCHOOLS SCIENCE COMPETITION

30 QUESTIONS
TIME ALLOWED: 45 MINUTES

STUDENT'S NAME: _____

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

Read the instructions on the **ANSWER SHEET** and fill in
your **NAME**, **SCHOOL YEAR**, **GENDER** and the **LANGUAGE**
YOU FIRST SPOKE.

Use a 2B or B pencil.
Do **NOT** use a biro or pen.
Rub out any mistakes completely.

Your answers **MUST** be recorded on the **ANSWER SHEET.**

Use the information provided to choose the **BEST** answer
from the possible options.

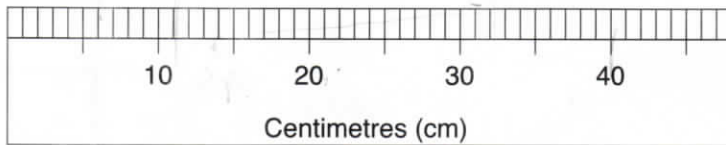
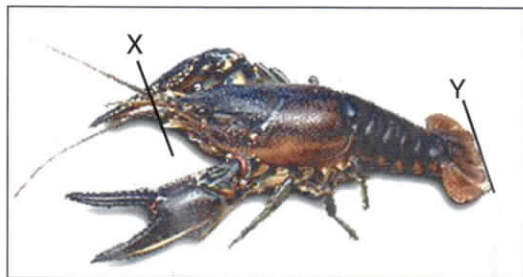
On your **ANSWER SHEET** blacken the oval that matches
the answer you choose.
Mark only **ONE** answer for each question.

Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

You may use a calculator and a ruler.

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1. This is a scale photograph of a freshwater lobster.

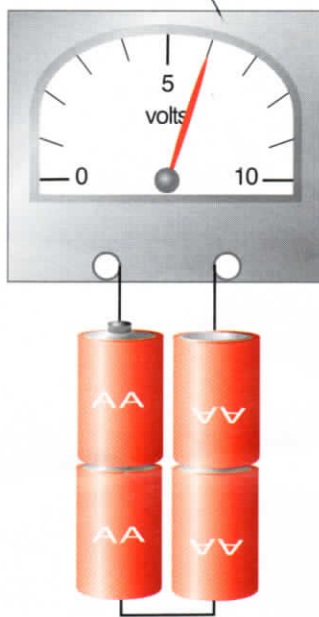
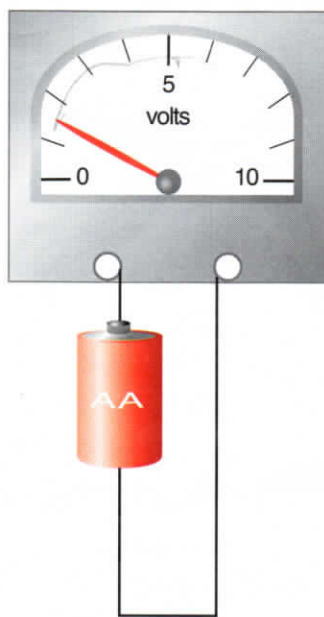


Using the scale provided, what is its body length from X to Y?

- (A) 10cm
- (B) 21cm
- (C) 33cm
- (D) 48cm

2. John used a meter to test the voltage produced by one battery and four batteries.

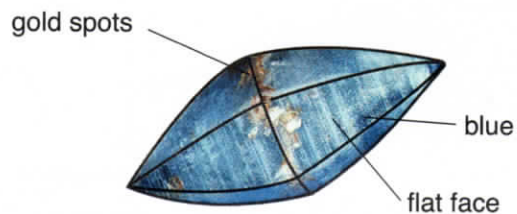
The diagrams show what he found.



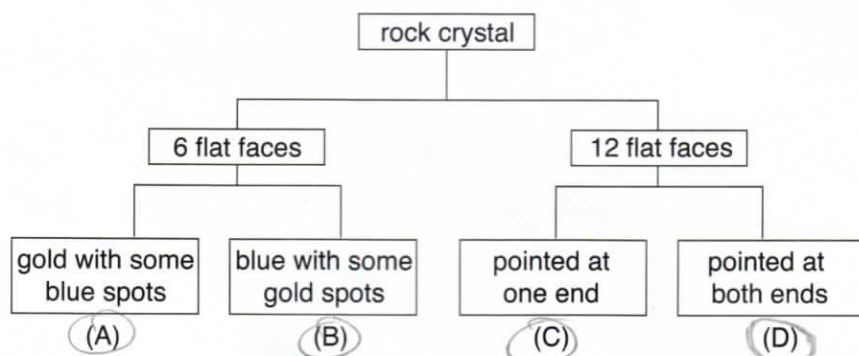
What is the difference in voltage as shown by the scales?

- (A) 0.5 volts
- (B) 1.5 volts
- (C) 4.5 volts
- (D) 6.0 volts

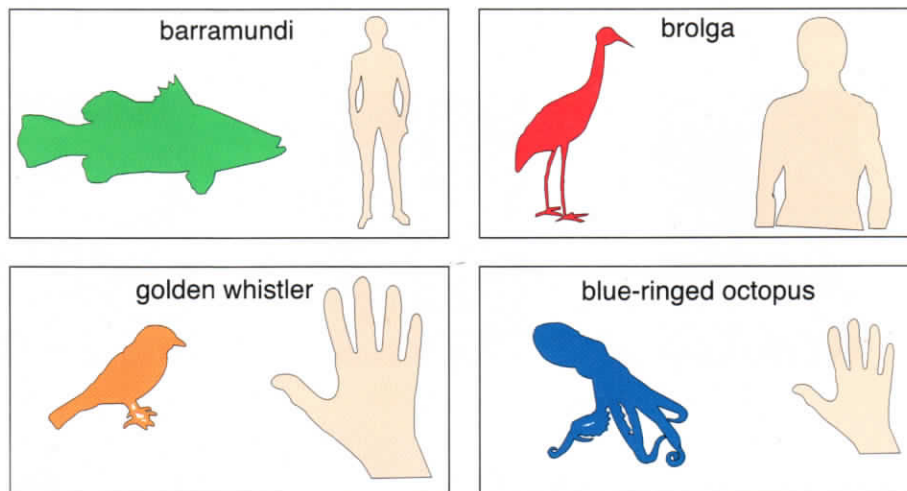
3. This photograph shows some of the faces of a rock crystal.



Use the chart below to find the letter that matches the rock crystal.



4. The sizes of these animals have been shown compared to adult human size.



Which of these animals is the smallest?

- (A) barramundi
- (B) blue-ringed octopus
- (C) brolga
- (D) golden whistler

5. Tony used this flow chart to group some animals.



Which animal belongs to group X?



© Drawings by Scott Camazine, <http://beelab.cas.psu.edu>

6. Fireworks are fired up into the sky where they explode into light then fall back to the ground.

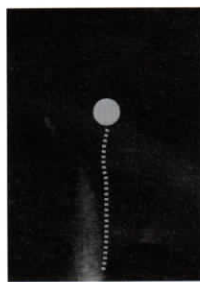
These pictures show what happened when a firework was fired. They are not in the correct order.



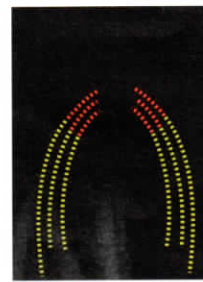
(P)



(Q)



(R)



(S)

Which list gives the correct order of what happened when the firework was fired?

- (A) R, S, P, Q
- (B) R, Q, P, S
- (C) P, Q, R, S
- (D) P, R, S, Q

7. Containers that have dangerous chemicals in them should show warning signs.

The pictures below show some of these signs.



Diane placed a label over the warning sign on a chemical container.

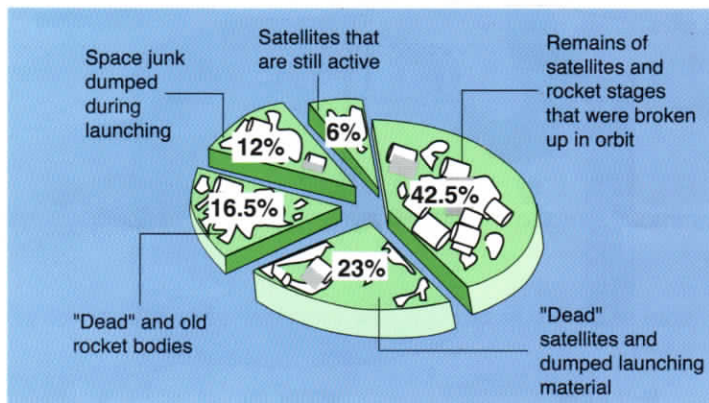


© Artwork courtesy of Seton Australia

What type of dangerous chemical was in the container?

- (A) flammable gas
- (B) flammable liquid
- (C) flammable solid
- (D) a chemical that is dangerous when wet

8. The diagram shows the percentages of man-made materials from different sources which are tracked as they orbit the Earth.



What percentage of these materials are still active?

- (A) 6%
(B) 12%
(C) 18%
(D) 48.5%

9. This is a drawing of a bee.



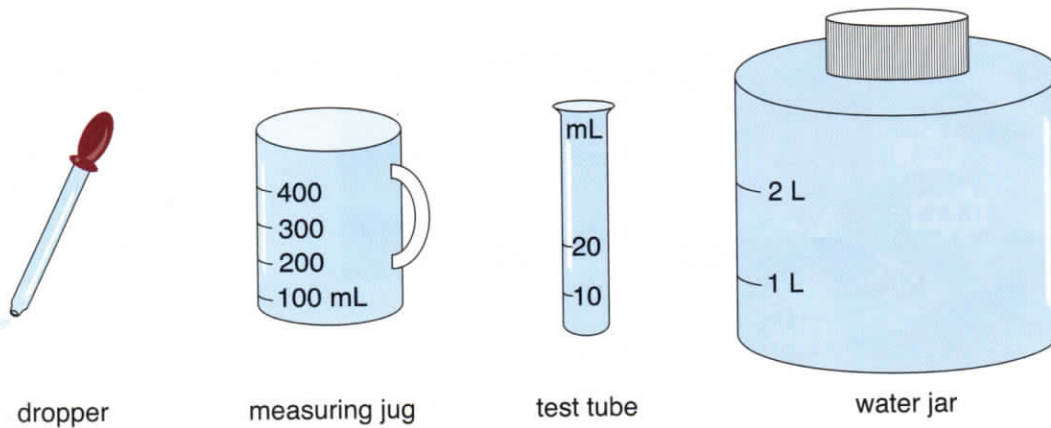
© Illustration by Scott Camazine, <http://beelab.cas.psu.edu>

To which part of the bee are its wings attached?

- (A) abdomen
(B) antenna
(C) head
(D) thorax

10. To test the water in a swimming pool, put 1 drop of chemical into 10 millilitres (mL) of water from the pool. Then shake the mixture and look at its colour.

These are pictures of some objects.

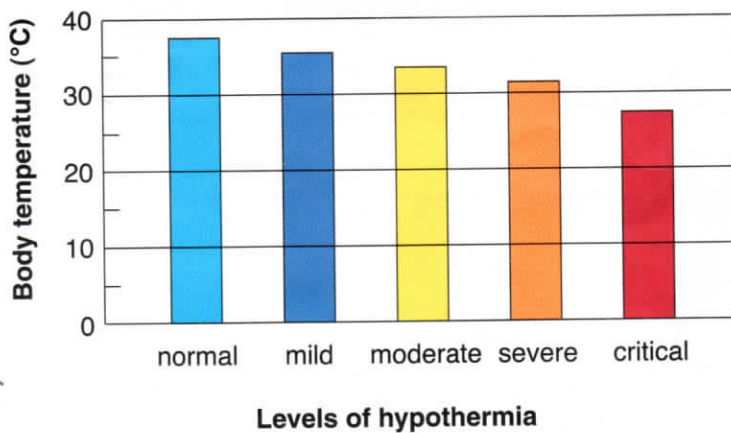


Which objects would best be used when testing water from a pool?

- (A) measuring jug and dropper only
- (B) measuring jug and test tube only
- (C) test tube and dropper only
- (D) test tube and water jar only

11. When a person's body temperature has dropped below normal levels, the person is said to be suffering from hypothermia.

The graph shows the body temperature for different levels of hypothermia.



A person was pulled unconscious from cold water. The person was found to have a body temperature of 32°C.

What level of hypothermia was this person suffering?

- (A) mild
- (B) moderate
- (C) severe
- (D) critical

12. The areas shaded on the maps show where four species of possums are found.



Eastern Pygmy Possum



Common Brushtail Possum



Common Ringtail Possum



Honey Possum

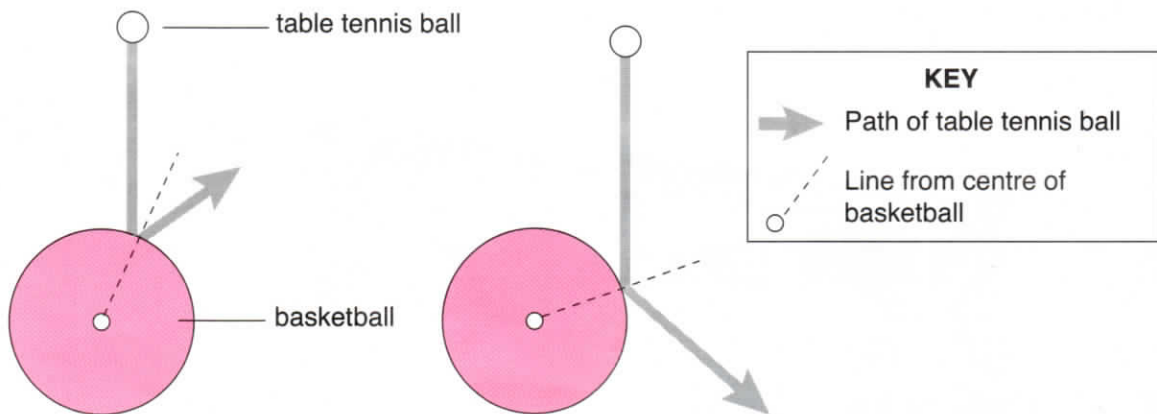
The same species of possum was sighted at the three locations X, Y and Z shown on the map.



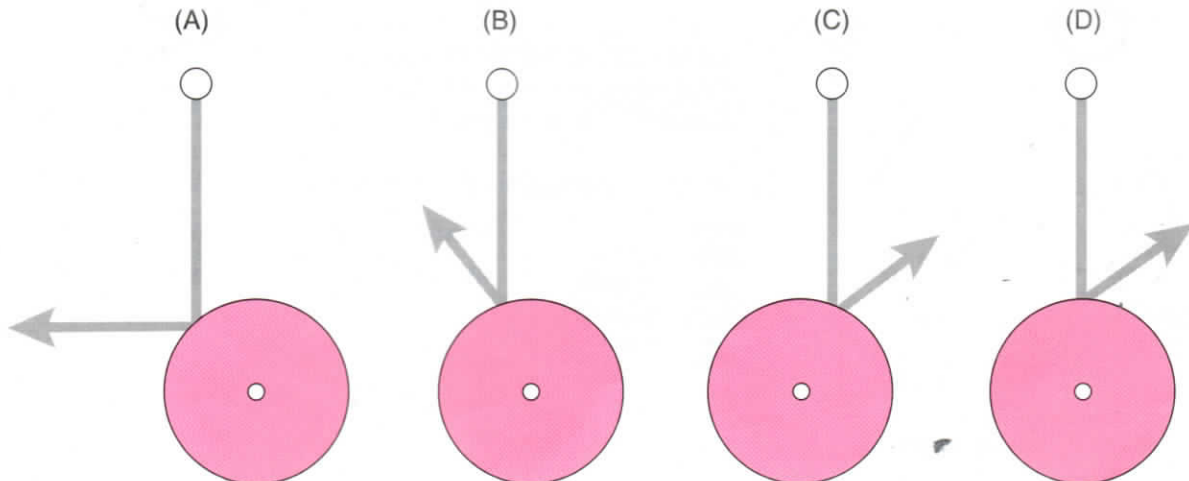
Which possum species was this?

- (A) Common Brushtail Possum
- (B) Common Ringtail Possum
- (C) Eastern Pygmy Possum
- (D) Honey Possum

13. When a table tennis ball is dropped on top of a basketball it will rebound as shown in the diagrams below.



Which diagram shows a rebound that is **unlikely** to occur?



14. Fish living in murky water usually have silvery skin.

Fish living in clear water near the bottom tend to be dark coloured.

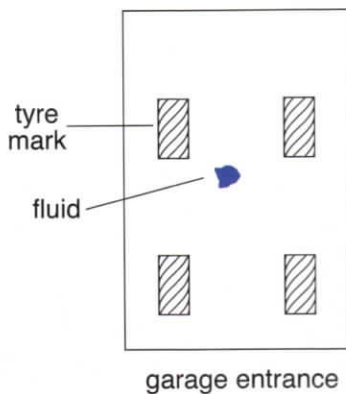
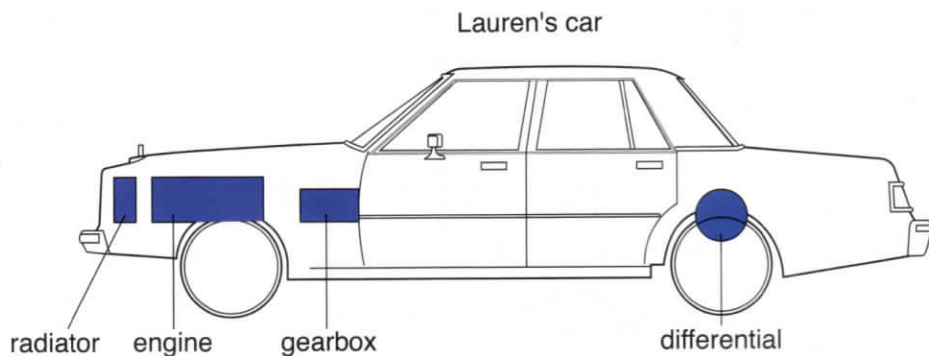
The table shows the type of fish that are most likely to be caught from the rocks on the coast.

Type of fish caught	Weather		Tide		Water	
	Hot	Cold	Low	High	Clear	Murky
bream	✓		✓	✓		✓
cod	✓		✓	✓	✓	
leatherjacket		✓		✓	✓	
salmon		✓	✓	✓		✓

Which dark coloured fish is likely to be caught in hot weather?

- (A) bream
(B) cod
(C) leatherjacket
(D) salmon

15. The diagram shows some parts of Lauren's car.

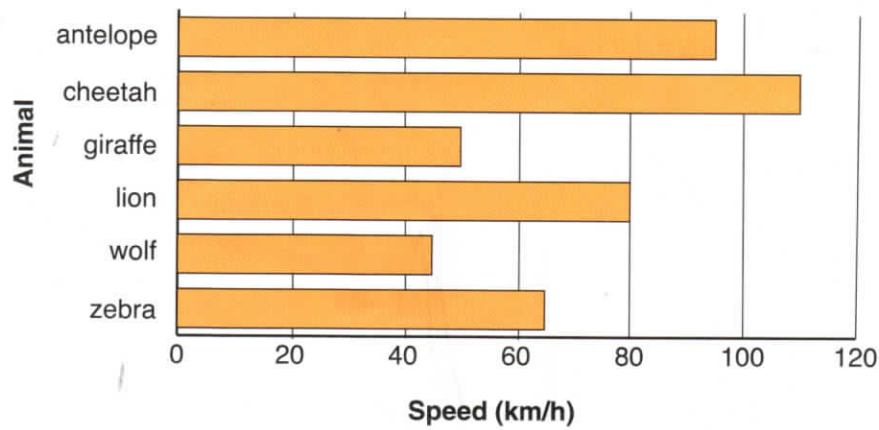


Lauren drives her car into her garage every evening, and reverses it out every morning. She noticed that fluid from her car had leaked onto the floor of the garage.

Which part of Lauren's car did the leak most likely come from?

- (A) radiator
(B) engine
(C) gearbox
(D) differential

16. The graph shows the speeds at which some animals can travel.

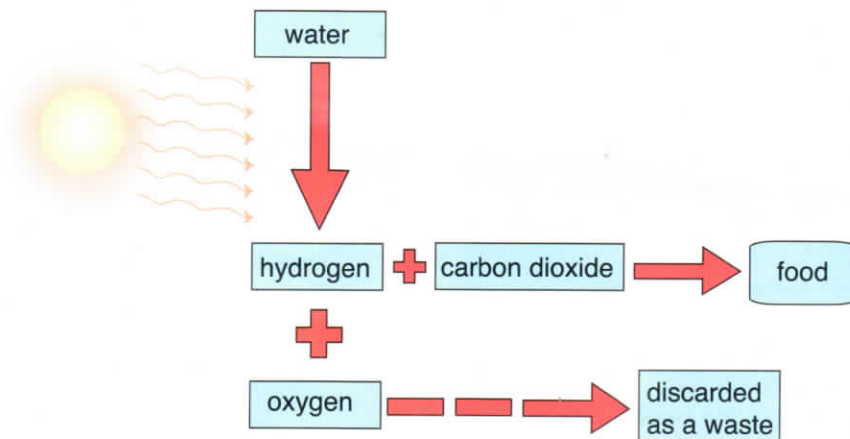


Which animal is faster than a wolf but slower than a zebra?

- (A) antelope
- (B) cheetah
- (C) giraffe
- (D) lion

17. Plants make their own food. This process is called photosynthesis.

The diagram shows what happens during photosynthesis.

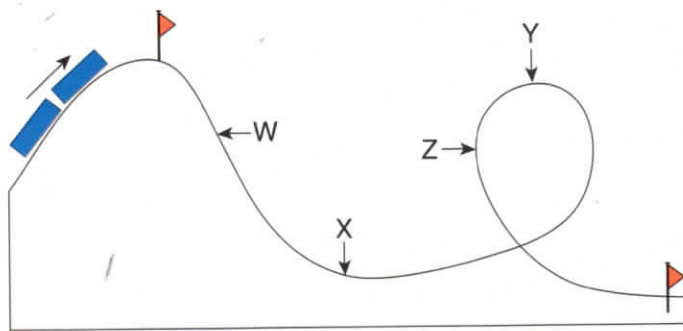


What do plants use to make food, and what is the waste of the process?

	Substance(s) used to make food	Substance(s) that are waste
(A)	hydrogen	oxygen
(B)	hydrogen	food
(C)	hydrogen and carbon dioxide	food
(D)	hydrogen and carbon dioxide	oxygen

18. A roller coaster changes stored energy into energy due to moving. It has the most stored energy at the highest point of the ride. It has the least stored energy at the lowest point of the ride.

The diagram shows a side view of a roller coaster ride.

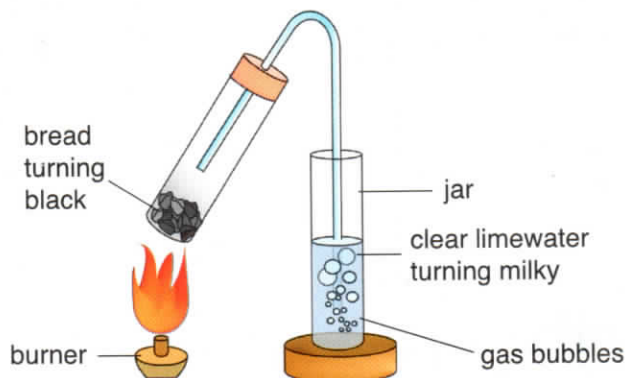


At which of the four points (W, X, Y and Z) along the ride would the roller coaster cars have the most stored energy?

- (A) W
- (B) X
- (C) Y
- (D) Z

19. An observation is information you can collect by seeing, hearing, smelling, feeling or tasting.

Mason made this drawing of his experiment.

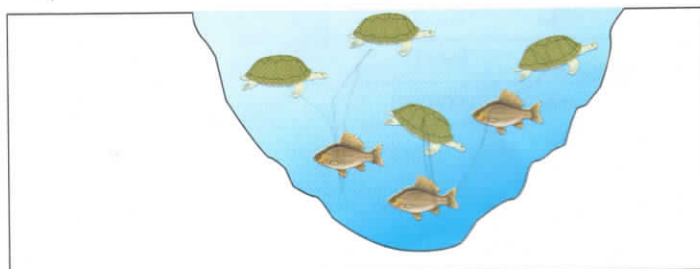


What observation could Mason make?

- (A) The clear limewater turned milky.
- (B) The bubbles were carbon dioxide gas.
- (C) Carbon dioxide gas came from the bread.
- (D) The bread turned black and gave off carbon dioxide.

20. Sharks and turtles eat fish for food. One turtle eats 50 fish to live. One shark eats 100 fish to live. Sharks do not eat turtles.

The picture below shows the number of fish and turtles in a lagoon.



KEY



= one turtle



= 100 fish

How many sharks could also live in the lagoon?

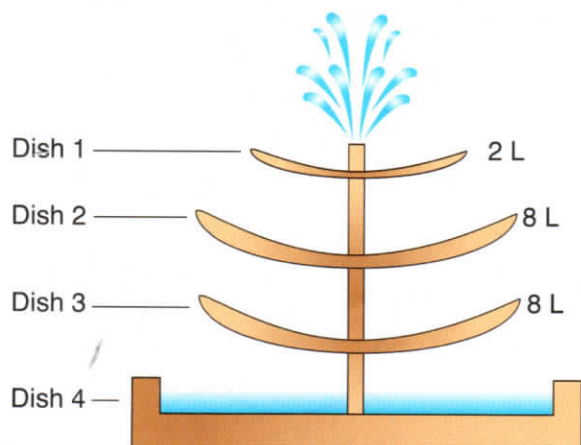
(A) 1

(B) 2

(C) 3

(D) 4

21. The diagram shows an empty fountain. The top dish (Dish 1) holds 2 litres of water. Dish 2 and Dish 3 both hold 8 litres of water.



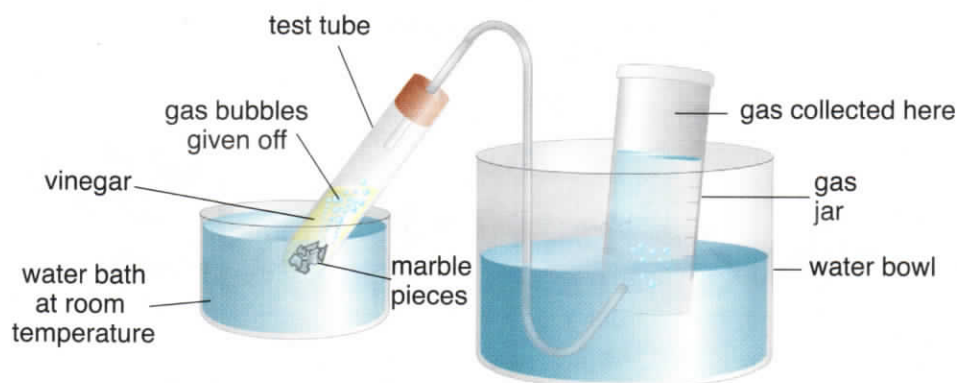
Water was pumped through a centre pipe from the bottom dish (Dish 4) to Dish 1 at a rate of 1 litre every 10 seconds.

How many seconds will have passed before water starts to flow into Dish 4?

- (A) 90 seconds
(B) 100 seconds
(C) 180 seconds
(D) 190 seconds

22. Rick wanted to investigate the reaction of marble with vinegar. He thought that the temperature at which the reaction took place and the size of the pieces of marble would affect the speed at which the marble reacted.

Rick did an experiment using pieces of marble of different sizes with the equipment shown. He measured the time taken to collect 10mL of gas when vinegar was added at room temperature (25°C). He repeated this experiment using small pieces of marble, and then repeated it again with powdered marble.



He then raised the temperature of the water bath to 50°C and performed all three tests again.

He recorded the following results.

Marble size	Time to collect 10mL of gas (minutes)	
	At 25°C	At 50°C
large pieces	10	9
small pieces	8	6
powdered	5	4

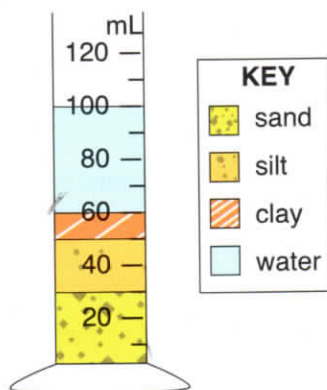
Which marble size reacted the fastest at which temperature?

	Marble size	Temperature (°C)
(A)	small	25
(B)	small	50
(C)	powdered	50
(D)	powdered	25

23. Tara wanted to know what fraction of a soil sample was sand.

She mixed some of the soil with water in a measuring cylinder and then let it stand overnight.

The result is shown in the diagram.



What fraction of the soil sample was sand?

- (A) 0.1
(B) 0.2
(C) 0.3
(D) 0.5

For questions 24 and 25 use the information below.

Sarah wanted to know how maggots came to be on bad meat.

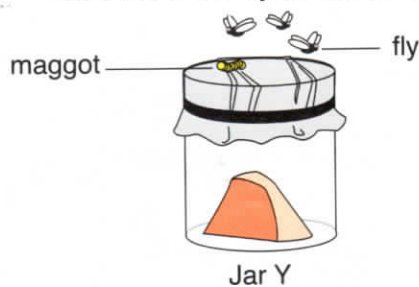
She set an experiment to test how that happened. The results after a few days are shown in the diagrams.

Jar sealed with plastic



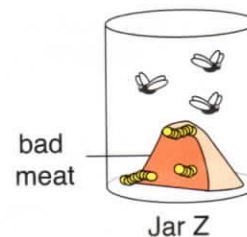
Jar X

Jar sealed with fly screen material



Jar Y

Jar open to the air



Jar Z

Sarah wanted her test to be fair.

24. What must be kept the same for each of the jars?

- (A) the material that was placed over the top of the jar
(B) the number of maggots on each piece of meat
(C) the type of meat used in each of the jars
(D) the number of flies that visited each jar

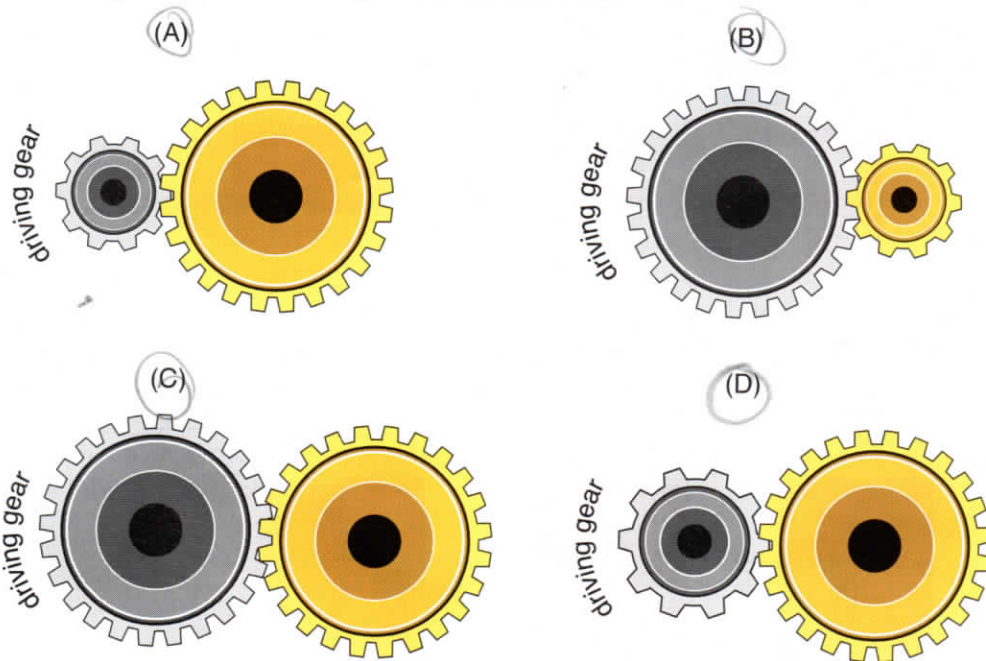
25. What were the conditions in Jar Y, and what were the conditions in Jar Z?

	Conditions in Jar Y	Conditions in Jar Z
(A)	Air cannot enter Flies can enter	Air cannot enter Flies cannot enter
(B)	Air can enter Flies cannot enter	Air can enter Flies can enter
(C)	Air can enter Flies cannot enter	Air can enter Flies cannot enter
(D)	Air cannot enter Flies can enter	Air can enter Flies cannot enter

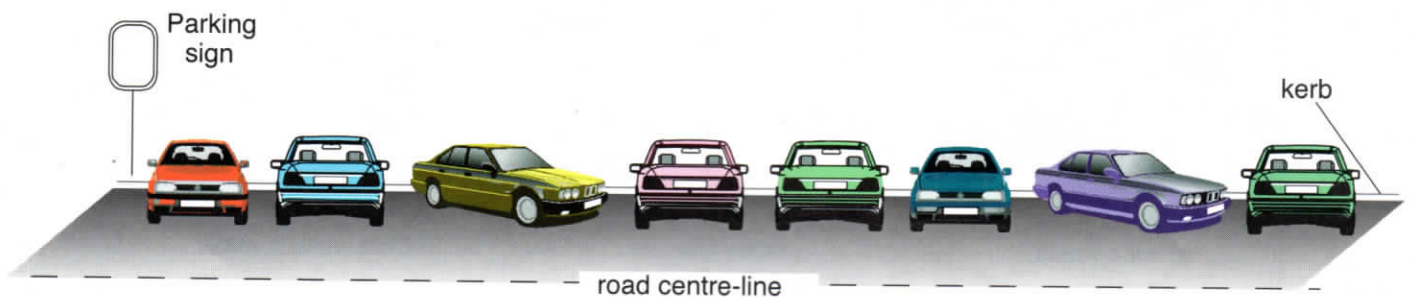
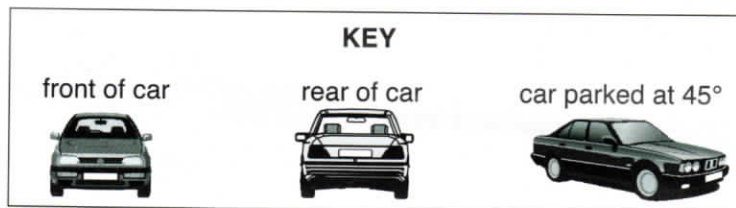
26. The following describes a set of gears in a machine:

"A motor turns a driving gear at 24 turns per minute. The driving gear is attached to another gear that only completes 10 turns a minute".

Which diagram represents the gears in this machine?

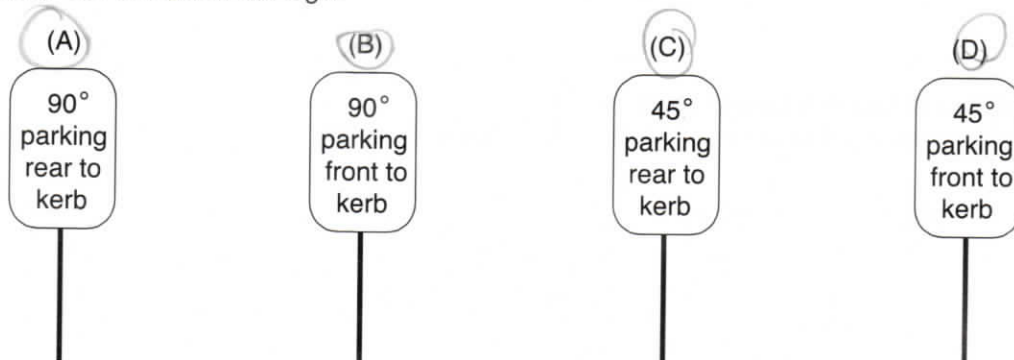


27. The diagram shows how some cars were parked.

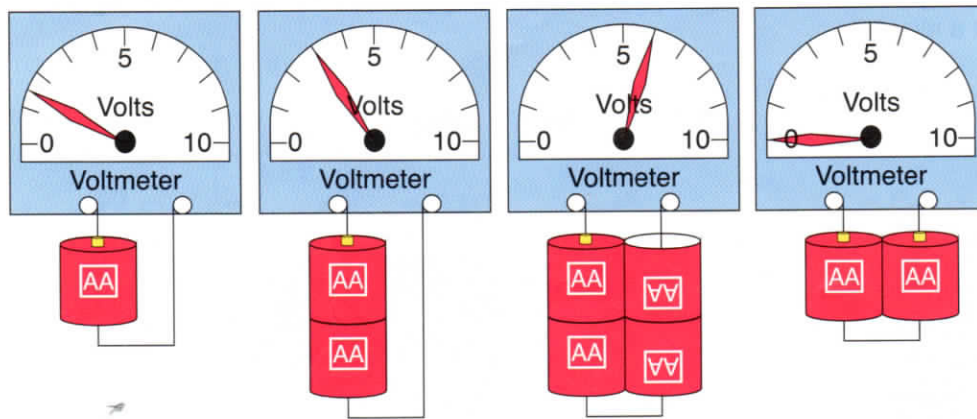


Four of the cars were **not** parked as the sign required them to. The sign had one instruction only.

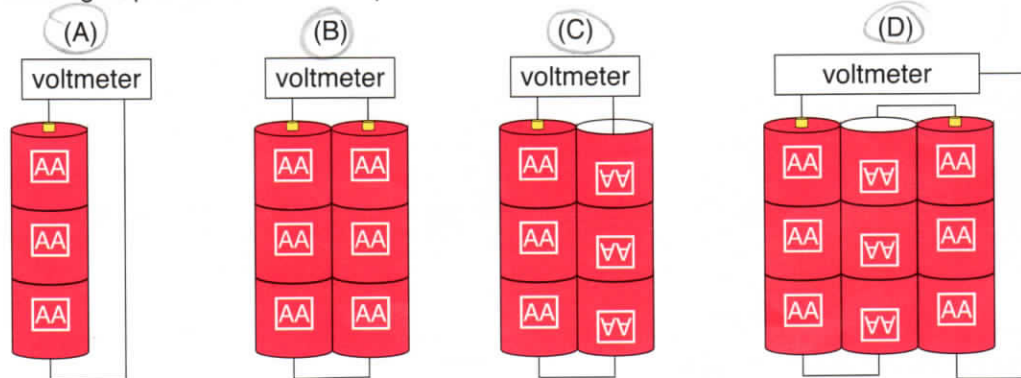
What was written on the sign?



28. John used a meter to test the voltage produced by batteries joined in different ways. The diagrams show what he found.



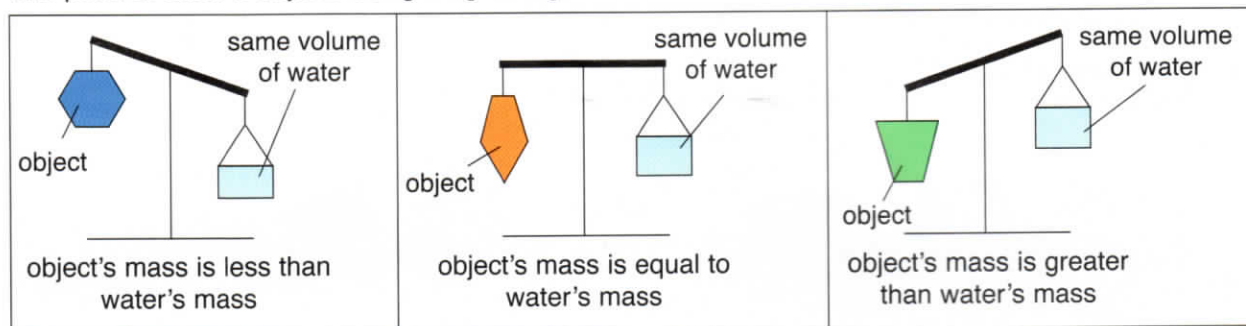
Which group of batteries would produce 9 volts on John's meter?



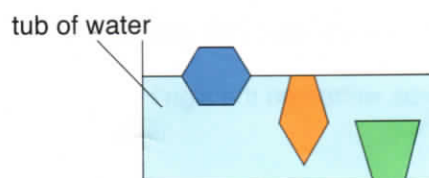
29. Some objects sink in water. Some objects float on water.

Objects with a mass greater than the mass of the same volume of water, sink in water.
Objects with a mass less than the mass of the same volume of water, float on water.

The pictures show 3 objects being weighed against the same volume of water.



This picture shows how the different objects sit in water.



Mel placed 4 different blocks in a tank of water.

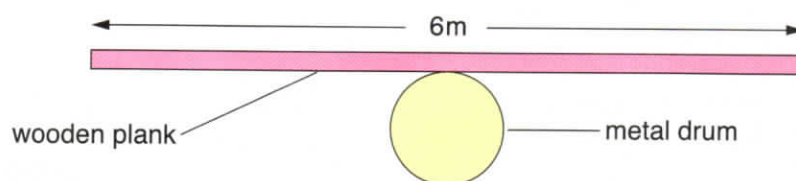
The mass of 1 block was equal to the mass of the same volume of water.

Which block was this?



30. Three children made a seesaw from a long plank of wood and a metal drum.

The seesaw is balanced with no children on it.



The children had different masses as shown.

John	30kg
Anne	20kg
Mary	10kg

Although the children had different masses, they found they could balance each other by changing where they sat on the plank.



The seesaw was balanced with Anne on one side and Mary on the other.

What distances were Anne and Mary from the centre of the metal drum?

	Anne	Mary
(A)	1m	2m
(B)	1m	3m
(C)	2m	1m
(D)	2m	3m

Yr 5 - ICAS SCIENCE

2002- Paper B

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- | | |
|---------|-------|
| 1. B | 16. C |
| 2. C | 17. D |
| 3. D | 18. C |
| 4. D | 19. A |
| 5. A | 20. A |
| 6. B | 21. C |
| 7. B | 22. C |
| 8. A | 23. D |
| 9. D | 24. C |
| 10. C | 25. B |
| 11. B/C | 26. A |
| 12. A | 27. B |
| 13. D | 28. C |
| 14. B | 29. B |
| 15. C | 30. A |

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